# PETTY EXPENSES MANAGEMENT SYSTEM

SRS Version: 1.3
Currency: INR (Whole Number)

Functional Specification: 3-Role Model (Admin, Maker, Checker), Shared Fund Accounts, Voucher Disbursements, Multi-Stage Approvals, Full Carry-Forward

Date: October 2026

## 1. Introduction & System Architecture

The Petty Expenses Management System is a web-based financial control platform engineered with a strict 3-role model (**Admin**, **Maker**, and **Checker**). The initial Administrator is created directly in the database / SQL seed, and holds complete authority to create, deactivate, reactivate, edit, and delete employee and administrator accounts. The system maintains a single shared fund balance for each assigned Maker-Checker pair. Every submitted expenditure passes through the independent Checker verification queue and receives a final decision from the Admin. The approval threshold is strictly Rs. 0 (no threshold auto-approvals). Financial amounts use whole-number INR values only (fixed integers, no paise).

## 2. User Roles & Segregation of Duties (3 Roles Only)

| Role Name | Core Scope & Responsibilities | Permissions & Boundary Limits |
|---|---|---|
| **Admin** | Primary System & Decision Authority. Initial Admin is seeded in SQL. Admin creates and manages all employee and admin accounts; assigns shared fund accounts; issues initial & supplementary vouchers; reviews & disburses fund requests; makes final Accept/Reject decisions on Checker-forwarded claims; views full audit trails. Can edit profile details and passwords for users in the directory modal. Can delete other admins and employees (self-deletion blocked). | Only Admin-accepted expenses post to financial ledger. Password changes require 6-digit email OTP; login does not require OTP. Cannot delete self. |
| **Maker** | Expense Preparation & Custodian. Operates assigned shared fund account. Prepares and submits expense claims (date, whole INR amount, category, payee, description, receipt). Submits additional fund requests when available balance is insufficient. | Cannot approve own claims. Cannot exceed available balance. Amount is reserved as an outstanding commitment immediately upon submission. |
| **Checker** | Internal Audit & Expense Verification Reviewer. Verifies supporting documents and available balance for every submitted claim. Exactly two actions: **Accept** (forwards to Admin) or **Send to Admin** (escalates with mandatory recorded comment). | Cannot make final posting decision. Cannot directly alter fund balances. Must review every expense regardless of amount. |

## 3. Business Rules Specification (BR-01 to BR-13)

| Rule ID | Rule Title | Rule Requirement & Functional Enforcement |
|---|---|---|
| **BR-01** | Voucher Disbursement | Every fund allocation requires a unique voucher (VCH-xxxx) containing beneficiary, whole INR amount, date, purpose, and payment details. |
| **BR-02** | Manual Payment Recording | Admin manually records payment details (mode, reference/UTR, date). Balance increases ONLY after voucher status is marked DISBURSED. |
| **BR-03** | Shared Fund Balance | All allocations, approved expenses, and adjustments record against one shared fund account for the assigned Maker and Checker. |
| **BR-04** | Rs. 0 Approval Threshold | Every expense requires Checker review and Admin decision regardless of amount. Threshold-based auto-approval is completely disabled. |
| **BR-05** | Checker Workflow | Checker has exactly two actions: 1) Accept (forward to Admin), or 2) Send to Admin (escalate with mandatory recorded comment). |
| **BR-06** | Admin Final Decision | Admin has two actions: 1) Accept (posts expenditure to ledger), or 2) Reject (mandatory reason; returns commitment to available balance). |
| **BR-07** | Available Balance Control | Expenditure cannot exceed Available Balance. Formula: Available Balance = Current Ledger Balance - Outstanding Commitments. |
| **BR-08** | Additional Fund Requests | Maker requests supplementary funds with mandatory reason. Admin approves/partially approves/rejects. Funds credit upon voucher disbursement. |

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|---|---|---|
| **BR-09** | Full Carry-Forward | Eligible closing balance carries forward in full as opening balance of subsequent month. No automatic expiry applies. |
| **BR-10** | Monthly Closing | System validates pending expenses, outstanding commitments, and reconciliation differences before formal period closure. |
| **BR-11** | Transaction Integrity | Posted transactions cannot be permanently deleted. Adjustments require authorized reversals with complete audit trail. |
| **BR-12** | Segregation of Duties | Maker prepares, Checker verifies, Admin decides and disburses. No single user can prepare and finalize an expense. |
| **BR-13** | Auditability & Logging | Every login, state change, voucher generation, approval, rejection, and profile edit records actor, timestamp, IP address, and details. |

## 4. Financial Ledger Equations & Balance State Transitions

- **Whole Number INR Values:** All amounts are strictly integer values (e.g. Rs. 1,200). Fractions and paise are prohibited.
- **Available Balance Equation:** Available Balance = Current Posted Ledger Balance - Outstanding Expense Commitments.
- **Submission Commitment:** When Maker submits claim for Rs. X, Available Balance -= X immediately.
- **Admin Acceptance:** Current Ledger Balance -= X (commitment converted to posted expenditure).
- **Admin Rejection / Cancellation:** Available Balance += X (commitment released back to available pool).
- **Disbursement Posting:** Current Ledger Balance += V and Available Balance += V upon voucher payment completion.
- **Monthly Closing Equation:** Closing Balance = Opening Balance + Total Disbursements - Total Admin-Accepted Expenses.

## 5. PostgreSQL Relational Database Schema (3 Roles)

| Table Name | Key Fields & Types | Constraints & Integrity Rules |
|---|---|---|
| **users** | id PK, employee_id UK, email UK, full_name, role (ADMIN\|MAKER\|CHECKER), designation, department, phone, photo, is_active, token_version | Auto-generated IDs (ADMxxxx, MKRxxxx, CHKxxxx). token_version revokes active JWTs upon password change. |
| **shared_fund_accounts** | id PK, account_id UK (FA-xxxx), name, maker FK, checker FK, admin FK, opening_balance, current_ledger_balance, available_balance | Single shared financial balance. Protects foreign keys on maker and checker. |
| **disbursement_vouchers** | id PK, voucher_number UK (VCH-xxxx), fund_account FK, voucher_type (INITIAL\|SUPPLEMENTARY), beneficiary, amount, payment_mode, payment_reference, payment_date, status, created_by FK | Voucher-based funding. Balances update only when status = DISBURSED. |
| **additional_fund_requests** | id PK, request_number UK (AFR-xxxx), fund_account FK, maker FK, requested_amount, approved_amount, reason, status, reviewed_by FK | Maker justification mandatory. Can trigger automated supplementary voucher upon Admin approval. |
| **expenses** | id PK, reference_number UK (EXP-xxxx), fund_account FK, maker FK, amount, category, payee, description, supporting_document, status, checker FK, checker_action, checker_comments, admin FK, admin_decision, admin_comments | States: SUBMITTED -> ACCEPTED_BY_CHECKER / SENT_TO_ADMIN -> ACCEPTED_BY_ADMIN / REJECTED_BY_ADMIN. Mandatory rejection reason. |
| **monthly_ledgers** | id PK, fund_account FK, year, month, opening_balance, total_disbursements, total_approved_expenditure, closing_balance, is_locked | Unique (fund_account, year, month). Closing balance automatically carries forward to next month. |
| **audit_logs** | id PK, user FK, action, record_type, record_id, details, ip_address, created_at | Immutable log tracking all security events, state transitions, disbursements, and decisions. |
| **otp_requests** | id PK, user FK, purpose, otp_code, expires_at, attempts, is_used | 6-digit numeric OTP, 5-minute expiry, max 5 attempts. Required for Change Password. Login does NOT require OTP. |

## 6. Production REST API Endpoints Matrix

| Method | Endpoint | Authorized Role | Functional Description |
|---|---|---|---|
| **POST** | /api/auth/login/ | Public | Authenticates via Email + Password -> returns JWT pair (NO OTP on login) |

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|---|---|---|---|
| **POST** | /api/auth/password/otp/request/ | Authenticated | Sends 6-digit email OTP for in-portal password change (5-min expiry) |
| **POST** | /api/auth/password/change/ | Authenticated | Verifies OTP & updates password, invalidating previous token sessions |
| **GET** | /api/dashboard/summary/ | All Roles | Role-specific metrics: Admin management queue, Maker balance, Checker verification queue |
| **GET/POST** | /api/admin/employees/ | Admin | Lists all users, filters by role/status, creates new Maker/Checker/Admin accounts |
| **GET/PATCH/DEL** | /api/admin/employees/{id}/ | Admin | View profile & claim history; PATCH name, designation, department, phone, role, password; DELETE user |
| **GET/POST** | /api/accounts/ | Admin | Lists all shared accounts or creates new account with initial disbursement voucher |
| **GET/POST** | /api/vouchers/ | Admin | Lists vouchers or issues supplementary voucher (credits ledger balance upon disbursement) |
| **POST** | /api/vouchers/{id}/disburse/ | Admin | Records manual payment mode, reference/UTR, and posts funds to account balance |
| **GET/POST** | /api/fund-requests/ | Maker | Lists fund requests or submits new request with mandatory business justification |
| **POST** | /api/fund-requests/{id}/review/ | Admin | Action: APPROVE, PARTIALLY_APPROVE, or REJECT. Optional instant voucher disbursement |
| **GET/POST** | /api/expenses/ | Maker | Lists expenses or submits new claim (reserves commitment from available balance) |
| **DELETE** | /api/expenses/{id}/ | Maker / Admin | Cancels a SUBMITTED expense and restores commitment back to available balance |
| **POST** | /api/expenses/{id}/checker-action/ | Checker | Action 1: ACCEPT (forwards to Admin) or Action 2: SEND_TO_ADMIN (mandatory comment) |
| **POST** | /api/expenses/{id}/admin-decision/ | Admin | Action 1: ACCEPT (posts to ledger) or Action 2: REJECT (mandatory remark; restores funds) |
| **GET/POST** | /api/ledgers/ | Admin | Reconciliation & period closing with automatic full carry-forward to next month |
| **GET** | /api/audit-logs/ | Admin | Searchable immutable audit logs for all transactions, disbursements, and approvals |